

CE 567: Highway Safety and Modeling

Course Code	CE 567
Course Name	Highway Safety and Modeling
L-T-P-C	3-0-0-3
Prerequisites	IC 252 Probability and Statistics; CE 352 – Transportation Engineering
Intended for	UG/PG elective
Mutual Exclusion	None
Approval	61st BoA (Jan 2026)

Preamble

This course provides comprehensive coverage of the state-of-the-art knowledge about highway safety and analysis of road safety data. This course covers aspects like collecting road safety-related data, integrating it, choosing appropriate analytical techniques, and evaluating countermeasures based on their efficacy. This course has been crafted after reviewing materials from various textbooks, research papers, and national and international codes of best practices. This course is immensely important for developing countries like India, as it can help achieve India's goal of reducing roadway fatalities in alignment with United Nations Sustainable Development Goal 3.6.

Course Modules

- **Module 1: Introduction to Road Safety:** Overview of road safety status in India and the world, development and function of road safety agencies, human factors, crash characteristics, crash data, roadway data, national road safety plans, road safety plans of different states in India, road safety audits, fundamentals of road safety [5 hours]
- **Module 2: Analytical Methods to Explore Road Safety:** Analyze Crash/Accident Data, Fundamentals of data collection, network screening, t-test, linear regression, injury severity analysis, crash frequency models [10 hours]
- **Module 3: Effectiveness of Crash Countermeasures:** Exploratory analysis of safety data, regression to mean, cross-sectional and panel studies, naïve before and after studies, empirical bayes method, crash modification factor, countermeasure selection, safety performance function [10 hours]
- **Module 4: Science of Crash Reconstruction:** Fundamentals of accident/crash studies, crash investigation procedures, causal factor analysis in crash events, use of vehicle dynamics, applications of reconstruction in legal, insurance, and safety contexts [5 hours]
- **Module 5: Crashes and Economics:** Economic appraisal, cost-benefit analysis, project prioritization, data-driven decision making, estimation of economic and societal cost of crashes [5 hours]
- **Module 6: Advances in Road Safety:** Surrogate safety measures, naturalistic driving, vehicle telematics data, V-box data, advanced driver assistance systems, driving simulator, autonomous emergency braking systems, use of machine learning and artificial intelligence in crash data analysis [5 hours]

Text Books

1. Lord, D., Qin, X., Geedipally, S.R., 2021. *Highway Safety Analytics and Modeling*. Science Direct. ISBN 978-0-12-816818-9
2. Washington, S., Karlaftis, M.G., Mannering, F., Anastasopoulos, 2020. *Statistical and Econometric Methods for Transportation Data Analysis*. Chapman and Hall/CRC, New York, USA.
3. Daily, J., Shigemura, N., Daily, J, 2018. *Fundamentals of Traffic Crash Reconstruction*. Institute of Police Technology and Management, Vol 3.

References

1. Indian Roads Congress, 1996, *Highway Safety Code*, Special Publication no. 44.
2. Indian Roads Congress, 2019, *Manual of Road Safety Audit*, Special Publication no. 88.
3. Indian Motor Vehicles (Amendment) Act, 2019.
4. Indian Roads Congress, 2018, *Guidelines for traffic calming measures in urban and rural areas*.
5. United States Department of Transportation, 2024, *Highway Safety Manual*
6. Ministry of Road Transport and Highways, 2010, *National Road Safety Policy*.